

Statistical Report on Industrial Defect Detection Datasets

July 7, 2026

1 Overview

This report gives a simple summary of the current dataset exploration work for the PatchCore study. At this stage, the main goal is not to train the model yet, but to first understand what kind of data we are working with.

The five selected industrial defect detection datasets are:

- MVTec AD
- VisA
- BTAD
- KolektorSDD2 (KSDD2)
- MVTec LOCO AD

For each dataset, I mainly checked the dataset structure, image categories, normal and defective samples, Ground-Truth masks, and defect size information. This helps us get a better idea of how different these datasets are before running PatchCore training and evaluation.

2 Current Outputs

The current output folder contains exploration results for the five datasets:

- MVTec AD_exploration
- visa_exploration
- btad_exploration
- ksdd2_exploration
- loco_exploration

Inside these folders, the generated results include several types of files. For example, there are category-level statistics, image-level defect size reports, category-level defect size summaries, dataset-level comparison charts, defect size charts, overlay images, and Excel summary files.

These outputs are useful because they show not only how many images each dataset has, but also how large or small the defect regions are.

3 Dataset-Level Statistics

The five datasets are quite different from each other. Some datasets contain many object categories, while others focus on only one or a few industrial surfaces. This means they may bring different levels of difficulty when used for anomaly detection experiments.

Note: The number of image-level defect records is based on the generated defect size reports. Each record usually corresponds to one defective image or one Ground-Truth mask used for defect area calculation.

Table 1: Basic Statistics of Five Industrial Defect Detection Datasets

Dataset	Number of Categories	Image-Level Defect Records	Notes
MVTec AD	15	1258	A classic industrial anomaly detection benchmark.
VisA	12	1200	A multi-category visual anomaly detection dataset.
BTAD	3	290	A smaller industrial defect dataset.
KolektorSDD2	1	356	A surface defect detection dataset.
MVTec LOCO AD	5	993	A dataset with both structural and logical anomalies.

4 Defect Size Calculation

For each defective image, the defect area is calculated from its Ground-Truth mask. The basic idea is simple: after converting the mask into a binary mask, we count how many pixels belong to the defect region.

The Defect Area Ratio is calculated as:

$$\text{Defect Area Ratio} = \frac{\text{Number of non-zero pixels in the Ground-Truth mask}}{\text{Full image area}} \quad (1)$$

The full image area is calculated as:

$$\text{Full Image Area} = \text{Image Width} \times \text{Image Height} \quad (2)$$

Then the Defect Area Percentage is:

$$\text{Defect Area Percentage} = \text{Defect Area Ratio} \times 100 \quad (3)$$

This percentage tells us how much of the whole image is covered by the defect. For example, a very small value means the defect is tiny and may be harder to detect or localize.

5 Visualization Results

The current analysis also includes several types of visualization results, these visualizations make the dataset statistics easier to understand. Instead of only reading numbers from CSV or Excel files, we can directly see how large the defects are and how they appear in the images.

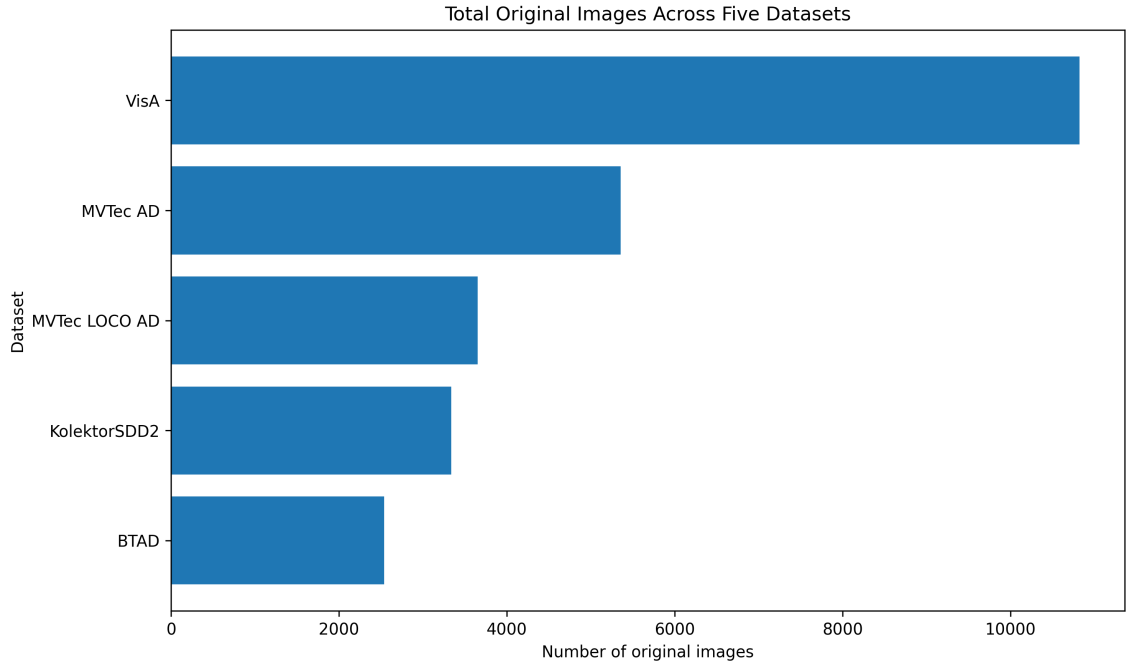


Figure 1: Comparison of total original images across the five datasets.

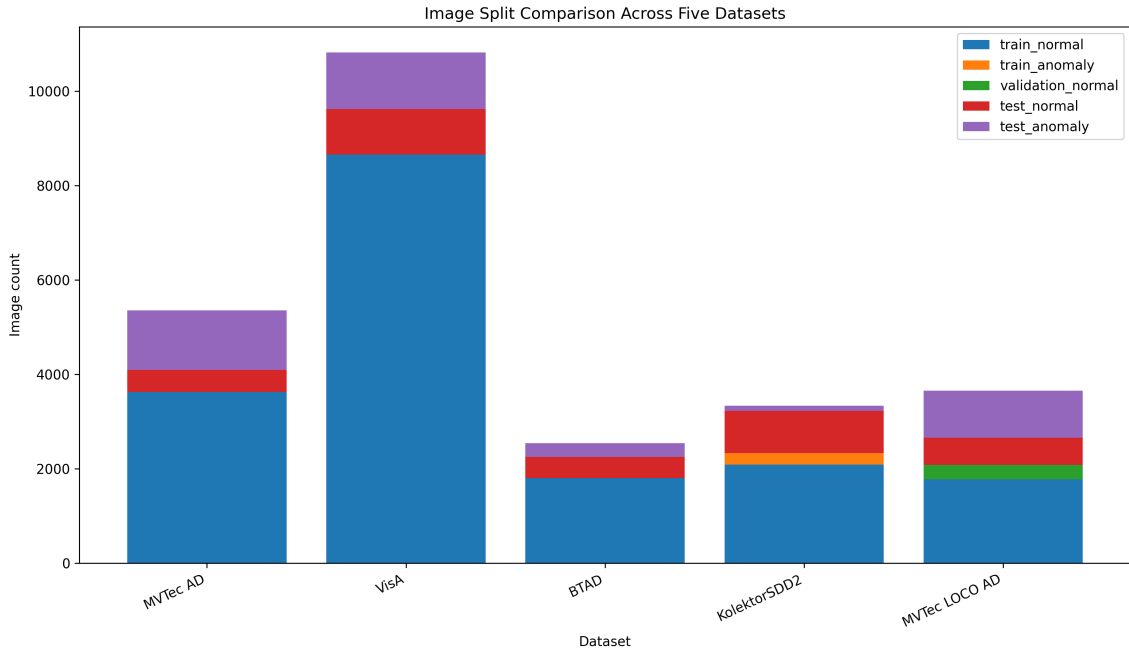


Figure 2: Comparison of image splits across the five datasets, including normal training images, normal test images, and abnormal test images.

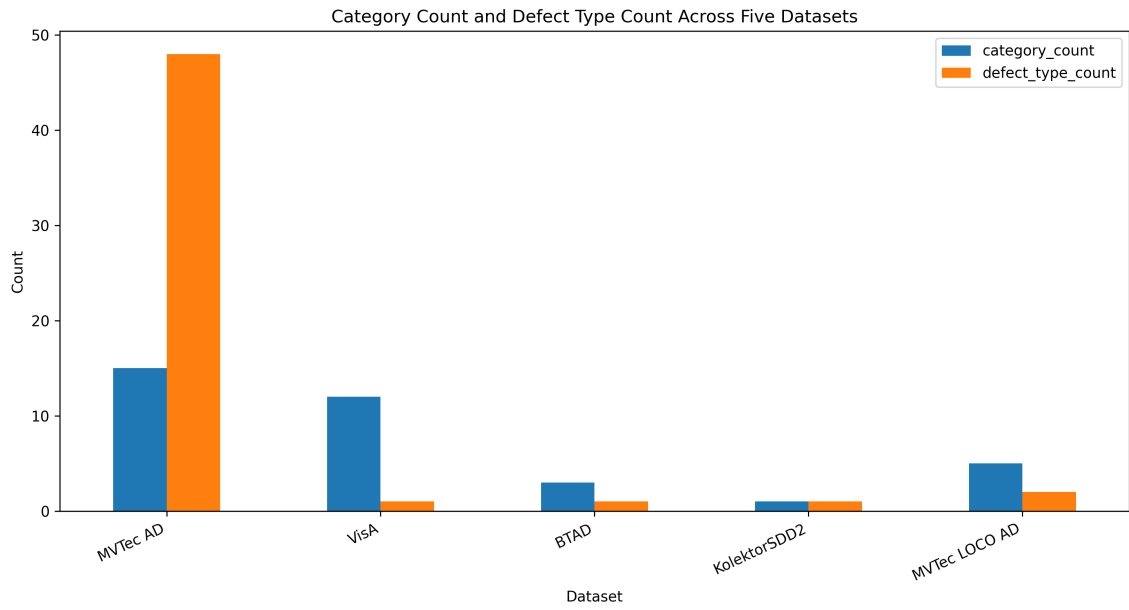


Figure 3: Comparison of category counts and defect type counts across the five datasets.

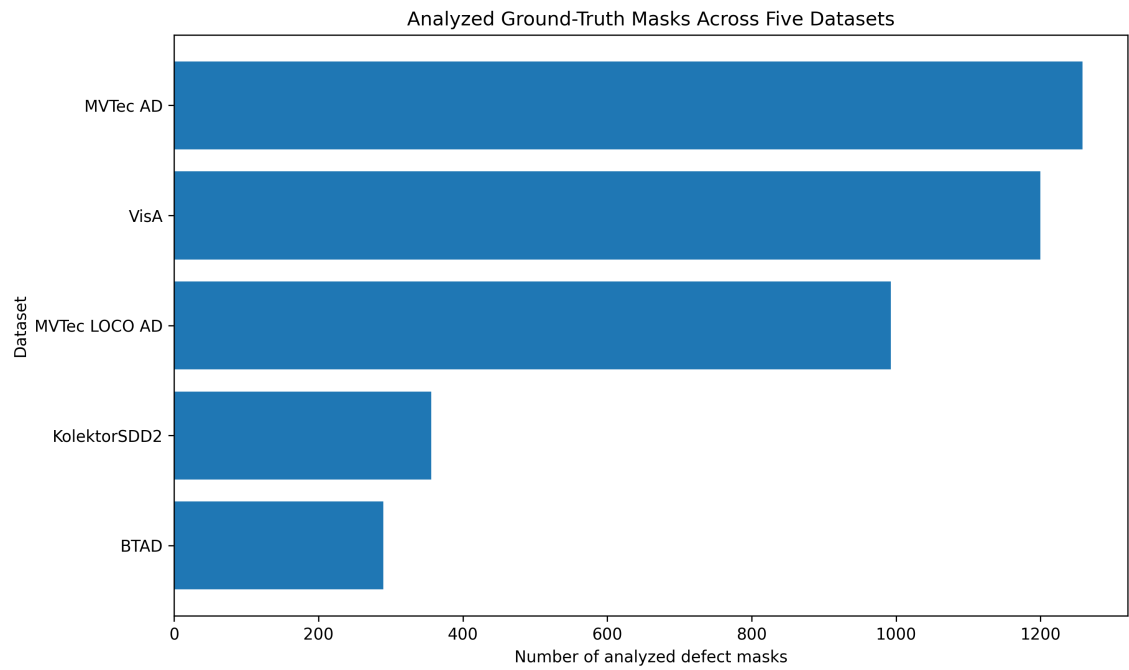


Figure 4: Comparison of analyzed Ground-Truth mask counts across the five datasets.

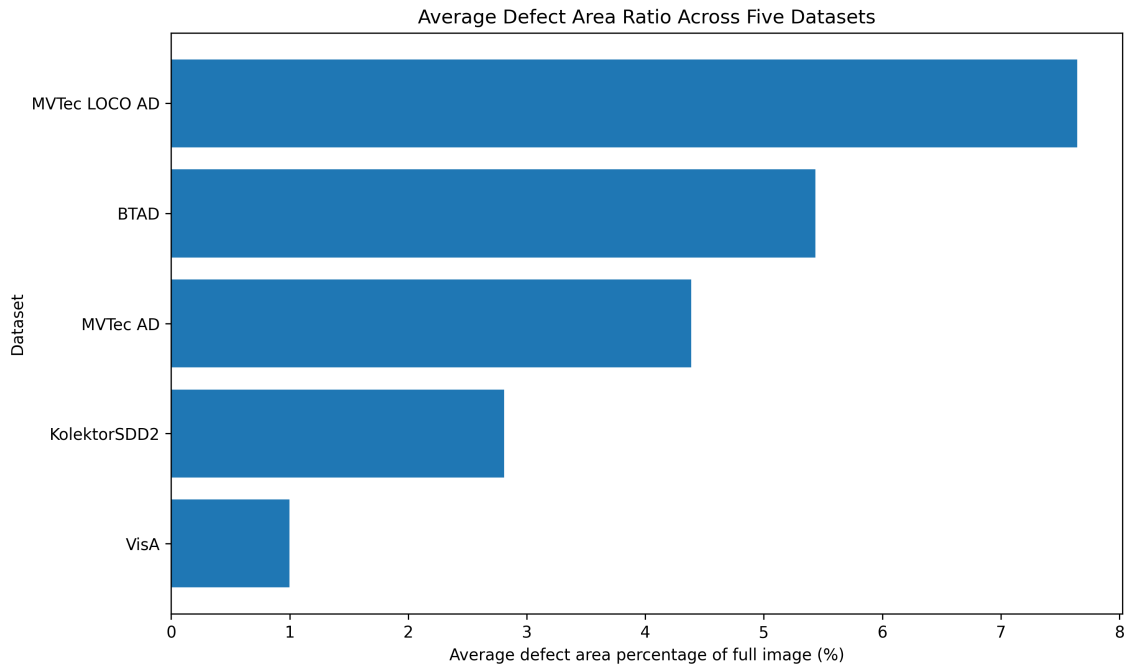


Figure 5: Comparison of average defect area percentage across the five datasets.

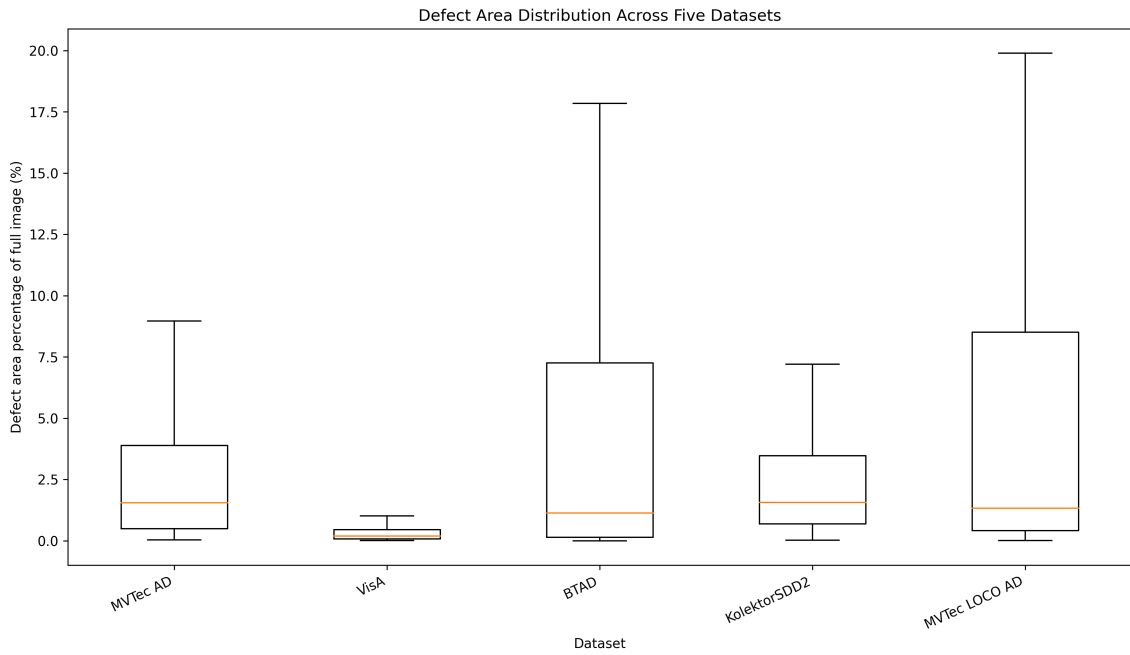


Figure 6: Boxplot comparison of defect area percentage across the five datasets.